



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

FINAL

APOLLO 13 FLIGHT PLAN

AS-508/CSM-109/LM-7

MARCH 16, 1970

FLIGHT PLANNING BRANCH
FLIGHT CREW SUPPORT DIVISION



MANNED SPACECRAFT CENTER
HOUSTON, TEXAS

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MCC-H

1313 CST

FLIGHT PLAN

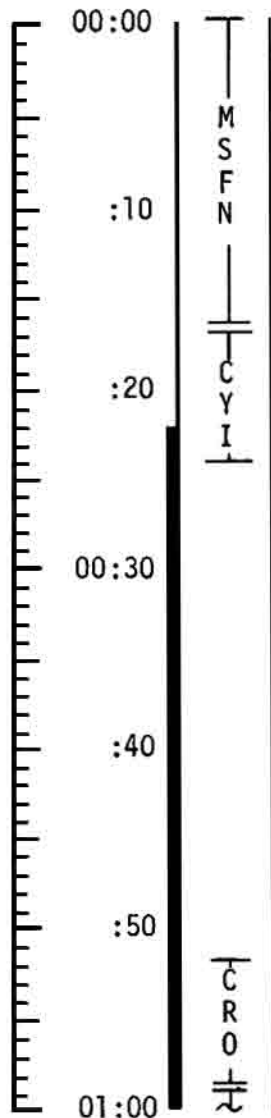
LIFT-OFF APRIL 11, 1970 LAUNCH OPERATIONS CHECKLIST

NOTES

LIFT-OFF CREW POSITIONS
LEFT COUCH - CDR
CENTER COUCH - CMP
RIGHT COUCH - LMP
AT SECO+20 SEC, S-IVB
MNVRS TO LH AND
INITIATES ORB RATE
(HEADS DOWN)

UPDATE TO CSM
Z TORQUING ANGLE

DUMP DSE



SECO
INSERTION AND SYSTEMS CHECKS

P52 - IMU REALIGN
OPTION 3 - REFSMMAT
(LAUNCH ORIENT)
GDC ALIGN

SET UP CAMERA EQUIPMENT

REPORT GYRO TORQUING ANGLES

P52 IMU REALIGN

N71: _ _ , _ _

N05: _ _ . _ _

N93: _ _ . _ _

X _ _ . _ _

Y _ _ . _ _

Z _ _ . _ _

GET _ _ : _ _ : _ _

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	00:00 - 01:00	1/E.O.	3-1

MCC-H

1413 CST

FLIGHT PLAN

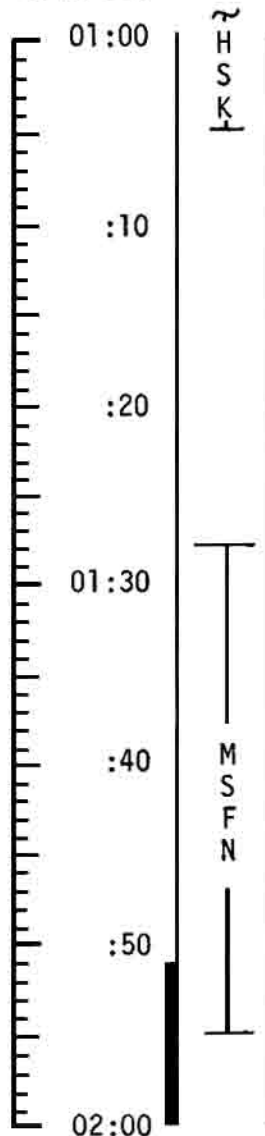
NOTES

DUMP DSE

UPLINK TO CM
CSM S.V. & V66
UPDATE TO CSM

TLI PAD
TLI +90 MIN
ABORT PAD
P37 (L/O+8) PAD

GO/NO-GO FOR PYRO
ARM



SCS ATT REF COMPARISON CK
EXTEND DOCKING PROBE

TV(MILA)

GO/NO-GO FOR PYRO ARM (CUE MSFN)
LOGIC ON

TLI PREPARATION CHECKLIST

GDC ALIGN
PYRO ARM

AS A GENERAL RULE,
EXCEPT DURING TEC,
UPLINK THE STATE
VECTOR TO THE CSM
SLOT AND TRANSFER
IT VIA V66 TO THE
LM SLOT IN ORDER TO
HAVE REDUNDANT STATE
VECTORS ONBOARD

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	01:00 - 02:00	1/E.O.	3-2

FLIGHT PLAN

TL1
BURN TABLE

P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
10°/SEC SHUTDOWN	+45° SHUTDOWN	BT + 6 SEC & V_i = PAD VALUE	NO TRIM

TABLE 3-1
3-2A

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MCC-H

1513 CST

FLIGHT PLAN

NOTES

GO/NO-GO

DUMP DSE

02:00

:10

:20

02:30

:40

:50

03:00

T
C
R
OM
S
F
NGO/NO-GO FOR TLI
TB-6 (02:25:49.65)

P47 - THRUST MONITOR

TLI

TIG: 02:35:27.65
BT: 5 MIN 55.7 SEC

P00 - CMC IDLING
 V66 - TRANS CSM SV TO LM SLOT
 TLI BURN STATUS REPORT
 CDR - TRANS TO CENTER COUCH, CMP - LEFT COUCH
 LMP - RIGHT COUCH
 WASTE STOWAGE VENT - CLOSED
 DIRECT O2 VLV-OPEN, UNTIL CABIN IS 5.7 PSI, THEN CLOSE
 GDC ALIGN
 SIVB MNVRS TO SEP ATT
 (2:56:25)

R 358 OMNI C
 P 140
 Y 320

AT SECO: S-IVB INERTIAL
 AT SECO+2 MIN 31 SEC:
 S-IVB TO LOCAL
 HORIZONTAL ORB RATE,
 HEADS DOWN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	02:00 - 03:00	1/TLC	3-3

MCC-H

1613 CST

FLIGHT PLAN

NOTES

GO/NO GO FOR T & D

03:00

:10

:20

03:30

:40

:50

04:00

M

S

F

N

T

V

ACTIVATE AND LOAD DAP N46 (11103, 11111)
LOAD DOCKING GIMBAL ANGLES
CSM SEP PREPARATION

CSM/SIVB SEP GET:03:06)

CSM MNVR TO DOCK ATT (03:10)
HGA TRACK - REACQ
HGA BEAM - WIDE DAP 11102
TV (GDS) 03:15 TO 04:23 CM4/TV - PEAK, BRKT (f 22)
VISUALLY INSPECT AND PHOTOGRAPH SIVB AND LM

R 302 HGA
P 320 P - 5
Y 040 Y 292

DOCK GET: 03:16)

BEGIN CSM/LM CABIN PRESSURE EQUALIZATION
CDR:CONFIGURE FOR LM EJECTION

TUNNEL PRESSURE INTEGRITY CHECK
REMOVE AND TEMPORARILY STOW TUNNEL HATCH
CHECK DOCKING LATCHES
VENT DOCKING PROBE
LM UMBILICAL CONNECTION
REINSTALL TUNNEL HATCH
LM TUNNEL VENT VLV - LM/CM ΔP
LEAVE TUNNEL EQUALIZATION VALVE CLOSED
CYCLE O₂ & H₂ FANS

S-IVB NON PROPULSIVE VENT START (GET: 03:41)

GO/NO-GO PYRO ARM (CUE MSFN)

LOGIC ON

LOAD DAP N46 (21101, 11111)

PYRO ARM

P47 - THRUST MONITOR

PHOTOGRAPH LM EJECTION

S-IVB VENT COMPLETE (GET: 03:56)

CSM/LM EJECTION

TIG: 04:01
BT: 3 SEC
ΔVR: 0.4 FPS
ULLAGE: N/A
ORBIT: N/A

T&D MNVR
+X FOR 3 SEC (ΔV ~ 0.5 FPS),
AFTER 15 SEC PITCH UP AT
2.0°/SEC. V49 AUTO MNVR
TO DOCKING ATT. NULL
TRANSLATION AND RATES,
+X FOR 4 SEC (ΔV ~ 0.7 FPS)

CAMERA SETTINGS FOR
TRANSPPOSITION/DOCKING:
CM2/DAC/18/CEX-BRKT,
MIR(f8,250,7)12fps,
0.7 MAG (5 MIN) MAG A
CM/EL/80 CEX - SPOT
(f8,250,FOCUS)10, MAG L

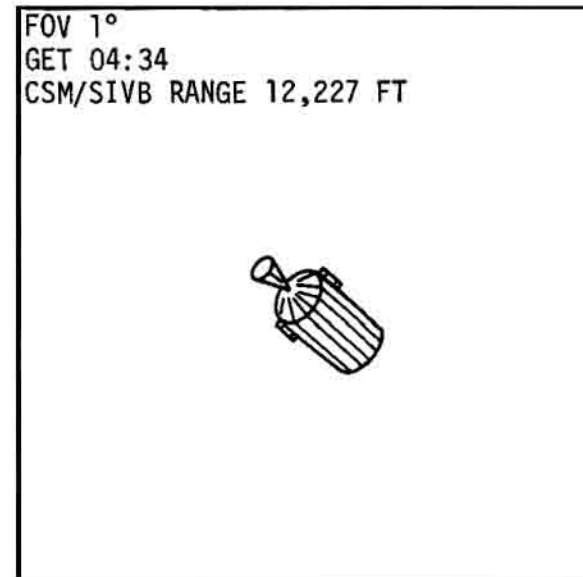
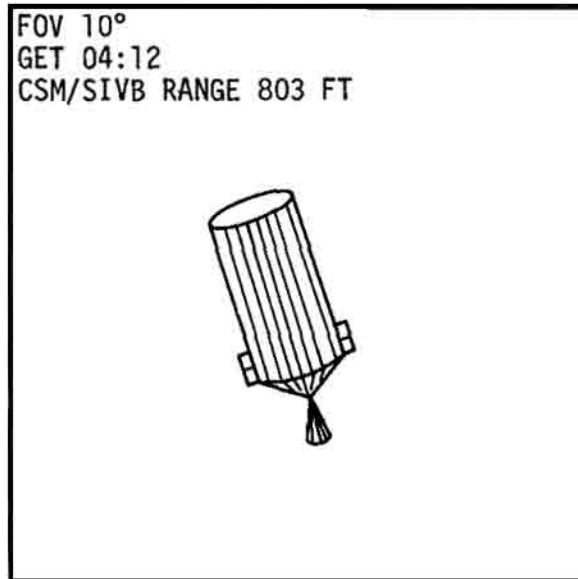
CAMERA SETTINGS FOR
LM EJECTION:
CM 2/DAC/18/CEX - BRKT,
MIR (f8,250,7) 6 fps,
0.3 MAG (5 MIN) MAG A

SPRING ACTUATOR
ΔV ~ 0.8 FPS. 5 SEC
AFTER EJECTION THERE
IS A 4 JET RCS -X
TRANSLATION FOR 3 SEC
(ΔV ~ 0.4 FPS). TOTAL
ΔV ~ 1.2 FPS.

GO/NO GO FOR
PYRO ARM AND
CSM/LM EJECTION

(TLI CUTOFF +
1HR 20 MIN)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	03:00 - 04:00	1/TLC	3-4



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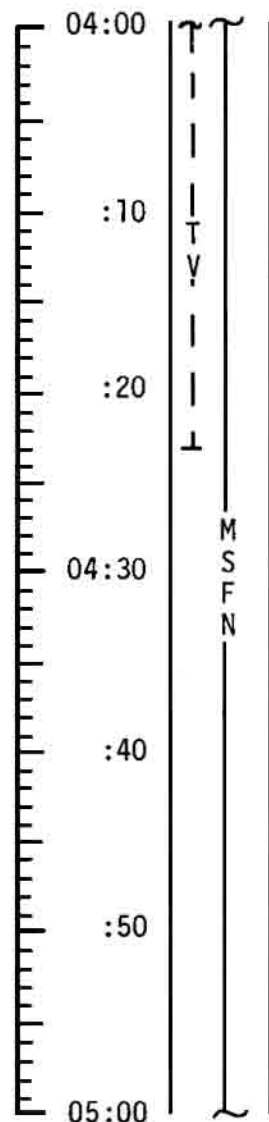
MCC-H

1713 CST

FLIGHT PLAN

NOTES

DUMP DSE



P00, V66 - TRANS CSM S.V. TO LM SLOT

MNVR TO ACQUIRE S-IVB IN HATCH

WINDOW (04:12)

R 094

OMNI D

P 323

Y 355

S-IVB APS EVASIVE MNVR

(GET = 04:14) (GROUND COMMAND)

 ΔV : 9.4 FPS

LOAD DAP, N46 (21111, 11111)

FLIGHT PLAN

BATTERY CHARGE, BATTERY B

S-IVB MNVRs TO PROPELLANT DUMP ATT (GET: 04:24)

MNVR TO P52 ATT

R 052

HGA P - 10

(04:25)

P 234

Y 350

Y 010

S-IVB CONTINUOUS H2 VENT ON (GET: 04:31)

S-IVB LOX DUMP (GET: 04:36)

HGA TRACK-MAN
HGA PITCH -52°
HGA YAW 270

DOFF & STOW
PGA's

TRANSFER ITEMS OUT
OF PGA POCKETS

THE S-IVB 80° YAW MNVR IS PLANNED FOR LM EJECTION +3 MIN. THE MNVR WILL NOT BE STARTED UNTIL CREW REPORTS GOOD EJECTION. THE S-IVB EVASIVE MNVR WILL NOT BE STARTED UNTIL THE CREW HAS THE S-IVB IN SIGHT.

NOTE: DURING TLC HGA IS REQUIRED ONLY FOR TD&E, TV TRANSMISSIONS AND MCC'S. THE ANTENNA WILL BE STOWED AT OTHER TIMES.

LOX DUMP ΔV
= 28.1 FPS

S/C INTERIOR PHOTOGRAPHY AT CREW OPTION

CM /DAC/5/CIN-SPOT (f2.8, 60, fixed) 6 fps, 1 mag MAG J

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	04:00 - 05:00	1/TLC	3-5

MCC-H

1813 CST

FLIGHT PLAN

NOTES

UPLINK TO CSM
DESIRED ORIENTATION
(PTC)
ZERO TRUNION BIAS

UPDATE TO CSM
P37 PAD (L/O+15)

05:00

:10

:20

05:30

:40

:50

06:00

M
S
F
N

P52 - IMU REALIGN
OPTION 1 - PREFERRED
(PTC ORIENT)
GYRO TORQUE

REPORT GYRO TORQUING ANGLES

GDC ALIGN

SECURE HGA, HGATRACK-MAN, HGA Pitch -52
HGA Yaw 270
VHF A SIMPLEX - OFF
VERIFY WASTE STOWAGE VENT VALVE - VENT

PREPARE FOR LAUNCH VEHICLE
SYSTEMS PERFORMANCE
DEBRIEFING AT 25 HRS.
SEE QUESTIONS ON PAGE 3-21

STARS _____

SA _____

TA _____

P52 IMU REALIGN

N71: _____

N05: _____

N93: _____

X _____

Y _____

Z _____

GET _____:_____:_____

P 37 PAD ASSUMES
NO MCC-1

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	05:00 - 06:00	1/TLC	3-6

MCC-H

1913 CST

FLIGHT PLAN

NOTES

06:00

MNVR TO OPTICS CALIBRATION ATT
 LOAD DAP, N46 (21101, 11111)
 P23 - Cislunar Navigation
 OPTICS CALIBRATION STAR 4 0
 P00

R 147
 P 340
 Y 0

OMNI B

S-IVB APS MCC-1
 GET: 06:00
 $\Delta V \sim 3$ FPS

:10

V49 - MNVR TO SIGHTING ATT
 STAR/EARTH HORIZON
 P23 - Cislunar Navigation

R 159
 P 297
 Y 0

~~OMNI~~

LOAD W MATRIX (R1 + 8 0 0 0 0) (R2 + 0 0 0 7 0)

3 MARKS ON EACH STAR
 INCORPORATE P23
 MARK DATA AND
 UPDATE ONBOARD
 STATE VECTOR

:20

1. STAR 33 EFH (R3 00120)

ANTARES

2. STAR 221 ENH (R3 00110)

DELTA CAPRICORNI

N88: (R1 +39954)(R2 -26599)(R3 -14003)

06:30

M
S
F
N3. STAR ¹²⁵~~33~~ EFH (R3 00120)^{EPS, LON}
~~STAR~~ SCORPIIN88: (R1 ¹²⁷³⁶~~-85957~~)(R2 ³⁷³²³~~-38386~~)(R3 ²⁸¹³³~~-34098~~)

:40

4. STAR 40 ENH (R3 00110)

ALTAIR

5. STAR 42 ENH (R3 00110)

PEACOCK

:50

O₂ FUEL CELL PURGE }
 WASTE WATER DUMP }

IF NO MCC-1

07:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	06:00 - 07:00	1/TLC	3-7

MCC-H

2013 CST

FLIGHT PLAN

NOTES

UPDATE TO CSM
QUADS TO DISABLE
FOR PTC (LOWEST
QUANTITY PRPLNT)

RECORD PHOTO
TIMES ON CREW
MARK.

07:00

:10

:20

07:30

:40

:50

08:00

M
S
F
N

MNVR TO PTC ATTITUDE
PREPARE FOR EARTH WEATHER

PHOTOGRAPHY
CM /EL/250/CEX - RING
(fTT,250,∞) 10

MAG L
FRAME

EAT PERIOD

P 90
Y 0

PHOTOGRAPH EARTH
1 FRAME EVERY 20
MIN FOR 3 HOURS.
MCC-H WILL RECORD
PRECISE
TIMES
FOR EACH
FRAME ON
MARK

EARTH
WEATHER
PHOTOGRAPHY

PTC
P 90 Y 0

GO TO PTC ATTITUDE
AND NULL RATES TO
+ 0.5° DB. DISABLE
TWO ADJACENT QUADS.
MONITOR FOR 20 MIN-
UTES THEN SELECT
+ 30° DB AND 0.3°/SEC
RATE, THEN DISABLE
THE REMAINING QUADS
AFTER SPIN UP.

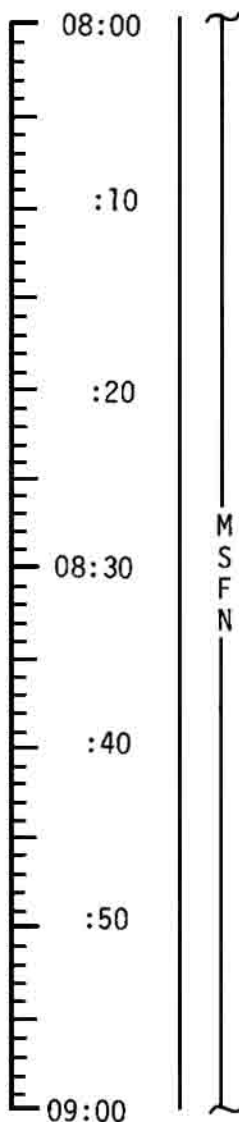
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	07:00 - 08:00	1/TLC	3-8

MCC-H

2113 CST

FLIGHT PLAN

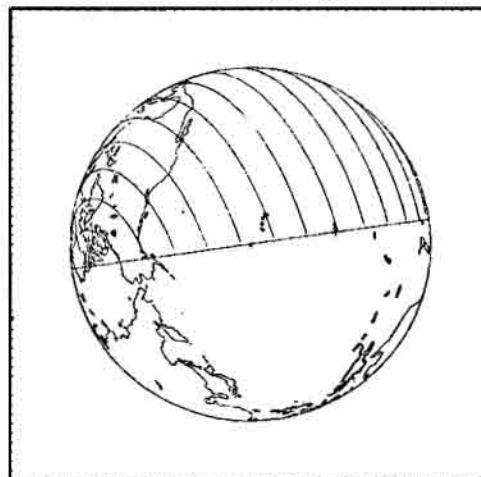
NOTES



DEACTIVATE PRIMARY EVAPORATOR
 GLY EVAP H2O FLOW - OFF (CTR)
 GLY EVAP STM PRESS AUTO - MAN
 GLY EVAP STM PRESS INCR - INCR FOR 1 MIN

SELECT NORMAL LUNAR COMM EXCEPT:
 S-BD AUX TAPE - OFF
 TAPE RCDR FWD - OFF

GET ¹⁰~~8~~ HRS F.O.V. 10°



LiOH CANISTER CHANGE
 (3 INTO A, STOW 1 IN B5)

EARTH
 WEATHER
 PHOTOGRAPHY
 PTC
 P 90 Y 0

S-IVB MCC-2 APPROX
 09:00 GET, ΔV IS
 NOMINALLY ZERO

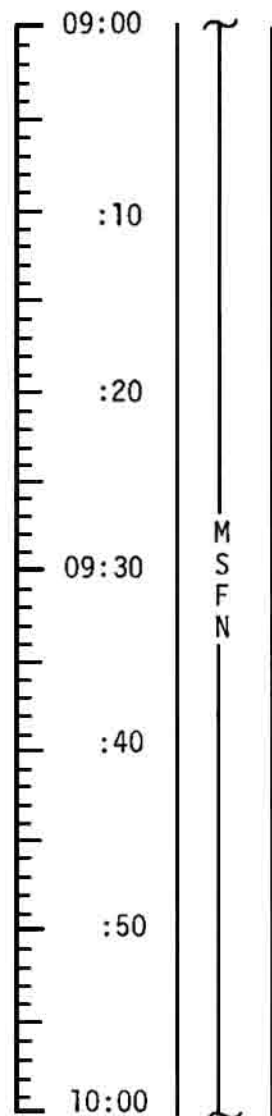
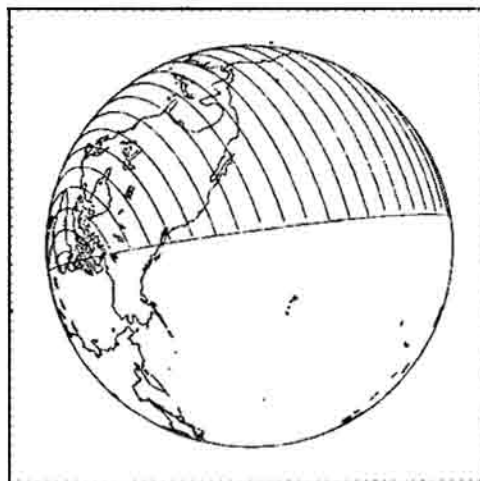
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	08:00 - 09:00	1/TLC	3-9

MCC-H

2213 CST

FLIGHT PLAN

NOTES

GET ⁸~~10~~ HRS F.O.V. 10°EARTH
WEATHER
PHOTOGRAPHYPTC
P 90 Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	09:00 - 10:00	1/TLC	3-10

MCC-H 2313 CST

FLIGHT PLAN

NOTES

UPLINK TO CSM
CSM S.V. & V66
MCC-1 TGT LOAD

UPDATE TO CSM
MCC-1 MNVR PAD

10:00
:10
:20
10:30
:40
:50
11:00

M
S
F
N

EMS ACCEL NULL BIAS TEST (REPORT)

CONTINUE EARTH WEATHER
PHOTOGRAPHY TO 11:30 IF MCC-1
IS NOT PERFORMED

P52 - IMU REALIGN
OPTION 3 REFSMMAT
(PTC ORIENT)

REPORT GYRO TORQUING ANGLES
CONTINUE PTC IF MCC-1 IS NOT PERFORMED

EARTH
WEATHER
PHOTOGRAPHY

PTC
P 90, Y 0

P52 IMU REALIGN
N71: ____'____
N05: ____'____
N93: ____
X ____.
Y ____.
Z ____.
GET ____:____:____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	10:00 - 11:00	1/TLC	3-11

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FLIGHT PLAN

MCC-1
BIIRN TABLE

P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	IF<2FPS, TRIM X AXIS TO 0.2FPS IF>2FPS, NO TRIM

TABLE 3-2
3-11A

0013 CST

FLIGHT PLAN

NOTES

11:00

:10

:20

11:30

:40

:50

12:00

MSFN

P30 - EXTERNAL ΔV

V49 - MNVR TO BURN ATT
P40/41 - SPS/RCS THRUST
SXT STAR CHECK

O₂ FUEL CELL PURGE } IF NOT PERFORMED
WASTE WATER DUMP } AT 06:55

GDC ALIGN

MCC-7

V66 - TRANSFER CSM SV TO LM SLOT
MCC-1 BURN STATUS REPORT

REPORT: LM/CM ΔP
WASTE STOWAGE VENT VLV - CLOSE (Δ8HRS FROM VENT)
VENT BATTS UNTIL SYSTEM TEST METER (4A) = 0

EARTH WEATHER
PHOTOGRAPHY

TIG: 11:41:25
BT: NOM. ZERO
ΔVR: NOM ZERO
ULLAGE: NONE
ORBIT: N/A

BURN STATUS REPORT

X	X		•		ΔTIG
X	X		•		BT
			•		V _{gx}
		TRIM		•	
X	X	X			R
X	X	X			P
X	X	X			Y
					V _{gx}
				•	V _{gy}
				•	V _{gz}
				•	ΔV _c *
X	X	X		•	FUEL *
X	X	X			OX *
X	X	X			UNBAL

*ITEMS TO BE
REPORTED TO MSFN

MCC-1 WILL BE
DELAYED TO MCC-2
IF PROPELLANT COST
IS NOT PROHIBITIVE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	11:00 - 12:00	1/TLC	3-12

0113 CST

FLIGHT PLAN

NOTES

MCC-H

UPDATE TO CSM
QUADS TO DISABLE
FOR PTC (LOWEST
QUANTITY PRPLNT)

UPDATE TO CSM
P37 PADS (L/O +
25, 35, 45 & 60)

12:00
:10
:20
12:30
:40
:50
13:00

M
S
F
N

MNVR TO PTC ATT P 90
Y 0

START PTC

EAT PERIOD

PRESLEEP CHECKLIST

PTC
P 90, Y 0

ONBOARD READOUT

BAT C _____
PYRO BAT A _____
PYRO BAT B _____
RCS A _____
B _____
C _____
D _____
DC IND SEL - MNA OR B

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	12:00 - 13:00	1/TLC	3-13

MCC-H

0213 CST

FLIGHT PLAN

NOTES

13:00
:20
:40
14:00
:20
:40
15:00

M
S
F
N

REST PERIOD
(10 HOURS)

PTC
P 90 Y 0

DURING REST PERIOD
TWO CREWMEN IN
REST STATIONS AND
ONE IN COUCH

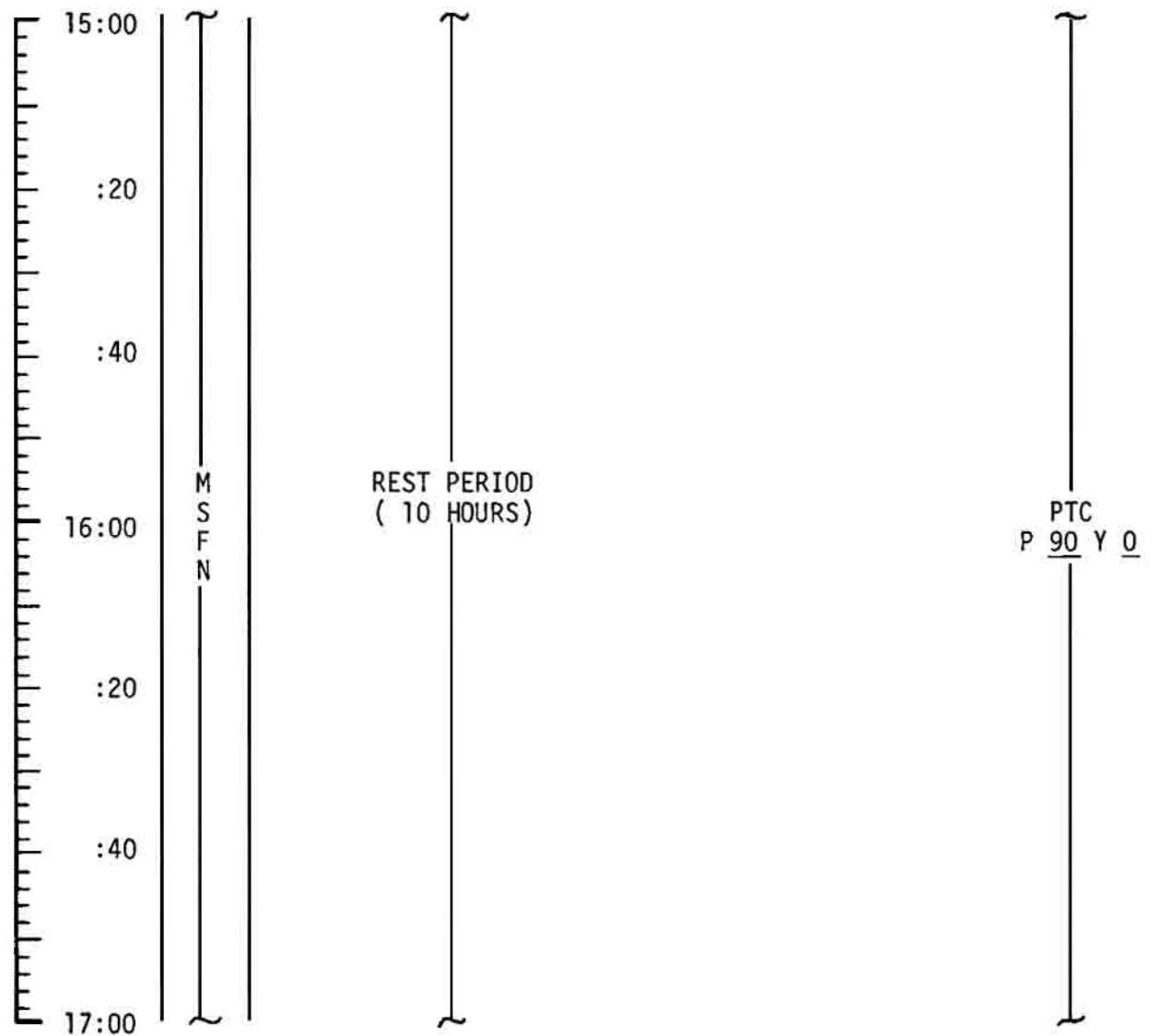
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	13:00 - 15:00	1/TLC	3-14

MCC-H

0413 CST

FLIGHT PLAN

NOTES



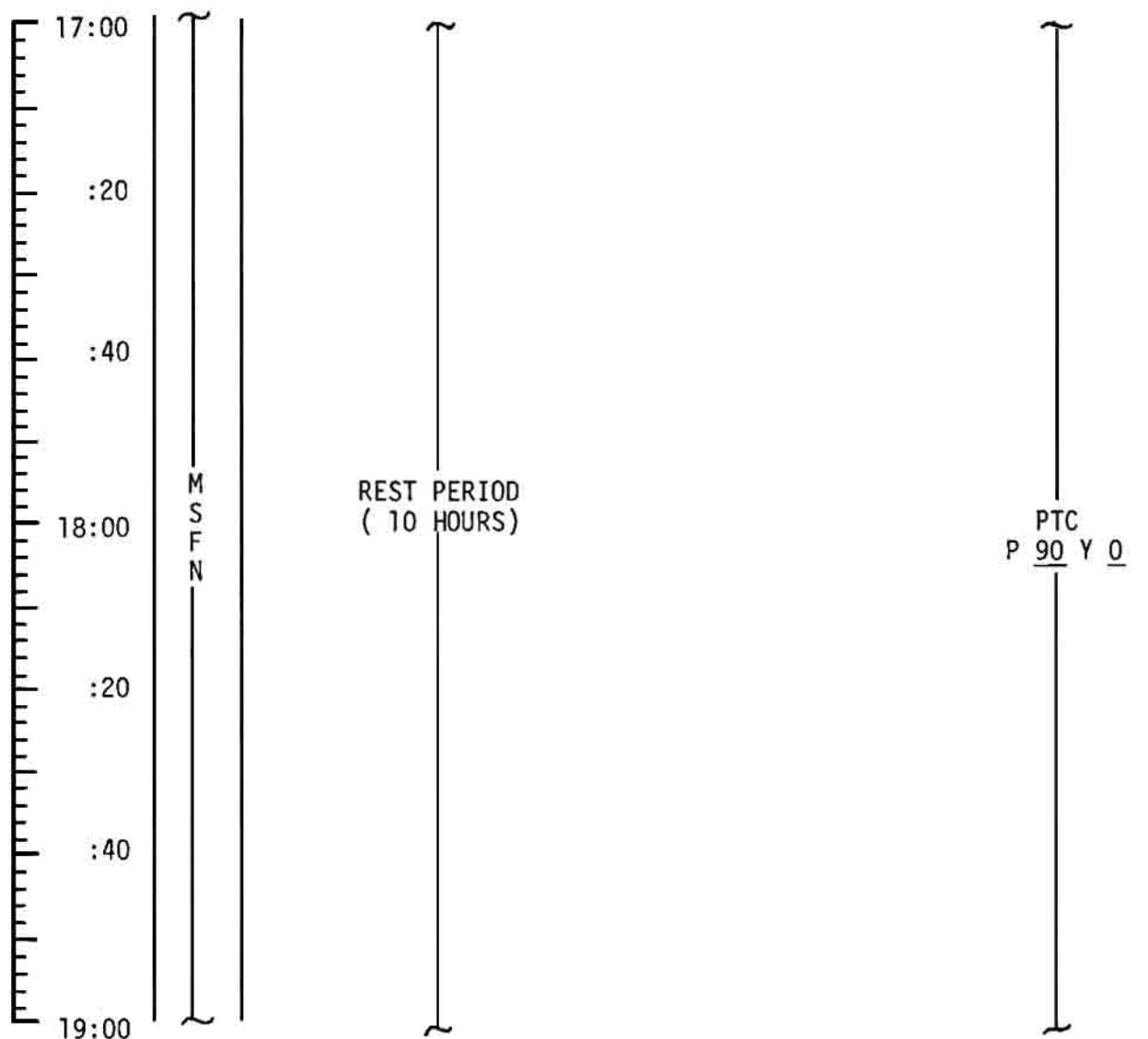
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APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	15:00 - 17:00	1/TLC	3-15

MCC-H

0613 CST

FLIGHT PLAN

NOTES



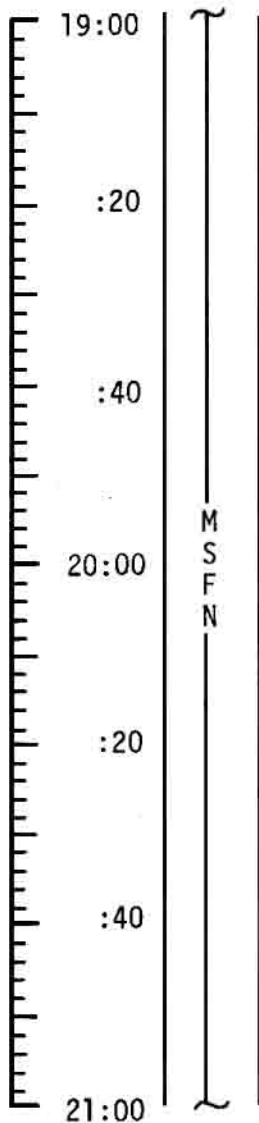
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	17:00 - 19:00	1/TLC	3-16

MCC-H

0813 CST

FLIGHT PLAN

NOTES



REST PERIOD
(10 HOURS)

PTC
P 90 Y 0

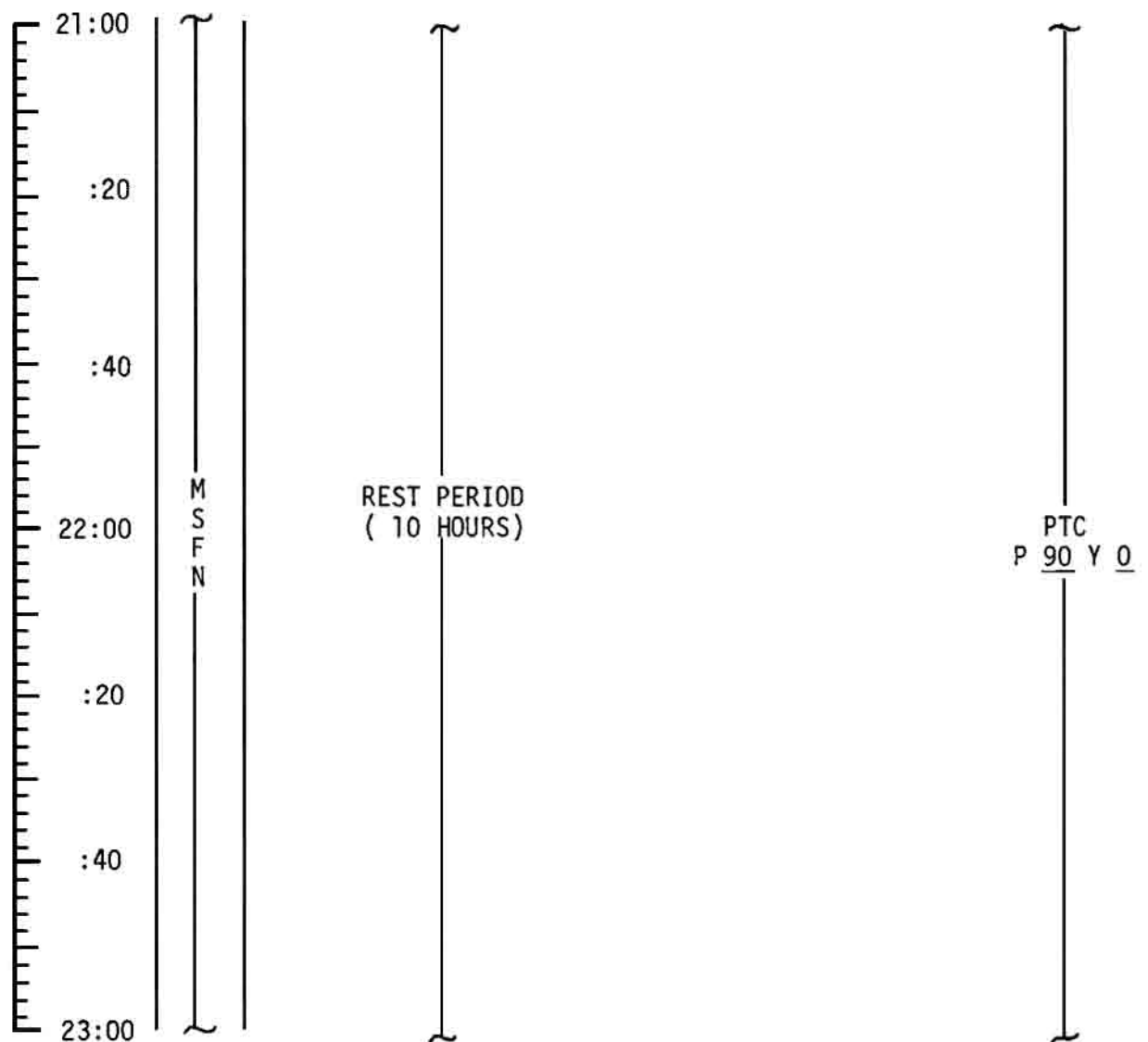
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	19:00 - 21:00	1/TLC	3-17

MCC-H

1013 CST

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	21:00 -23:00	1/TLC	3-18

1213 CST

FLIGHT PLAN

NOTES

MCC-H

UPDATE TO CSM
CONSUMABLES
FLIGHT PLAN

23:00
:10
:20
23:30
:40
:50
24:00

M
S
F
N

POSTSLEEP CHECKLIST

LIQH CANISTER CHANGE
(4 INTO B, STOW 2 IN B5)

BATTERY CHARGE, BATTERY A

EAT PERIOD

REPORT LM/CM ΔP

CSM CONSUMABLES UPDATE

GET: ____:____

RCS TOTAL ____

QUAD A ____ B ____

C ____ D ____

H₂ TOTAL ____

O₂ TOTAL ____

PTC

P 90, Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	23:00 - 24:00	2/TLC	3-19

MCC-H

1313 CST

FLIGHT PLAN

NOTES

24:00

:10

:20

:30

:40

:50

25:00

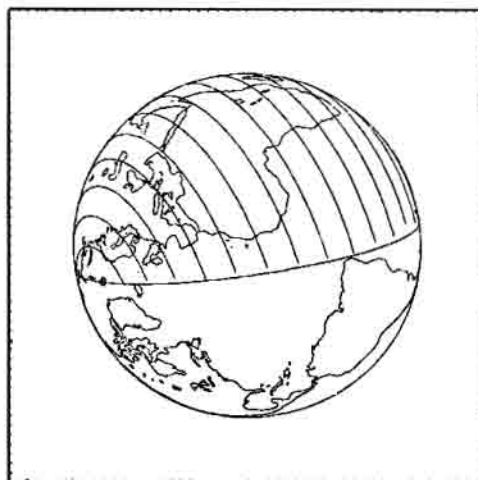
M
S
F
N

P52 IMU REALIGN
OPTION 3 REFSMMAT
(PTC ORIENT)

REPORT GYRO TORQUING ANGLES

GDC ALIGN

GET 25 HRS F.O.V. 5°



P52 IMU REALIGN

N71: __ __, __ __

N05: __ __, __ __

N93: __ __, __ __

X __ __, __ __

Y __ __, __ __

Z __ __, __ __

GET __ __: __ __: __ __

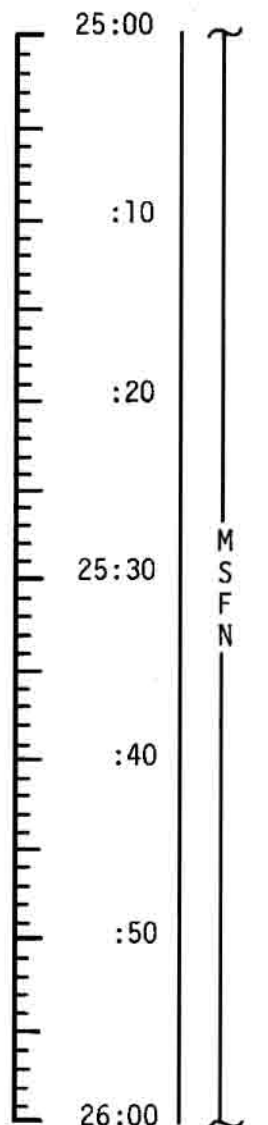
PTC
P 90 Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	24:00 - 25:00	2/TLC	3-20

MCC-HLAUNCH VEHICLE
DEBRIEFING

1413 CST

FLIGHT PLAN

NOTESLAUNCH VEHICLE SYSTEMS
PERFORMANCE DEBRIEFING

1. Were there any significant changes in noise level between stages of powered flight?
2. Were there any significant changes in noise/vibration level during a single stage of powered flight?
3. Were there any unexpected acceleration transients experienced at initiation of IGN, SII CECO, MAX Q or M/R shift for both SII and S-IVB?
4. After SC separation, describe the conditions of the IU thermal shroud. Was there any looseness?
5. How was ground/SC comm at ignition/liftoff time region relative to vibration and acoustic environments?
6. Describe any visible venting or suspected leak after separation.
7. When, and at what distance, was the S-IVB seen for the last time?
8. Are there any comments relative to S-IVB/IU TLI guidance cutoff conditions (predicted vs actual SC display)?

PTC
90 Y 0

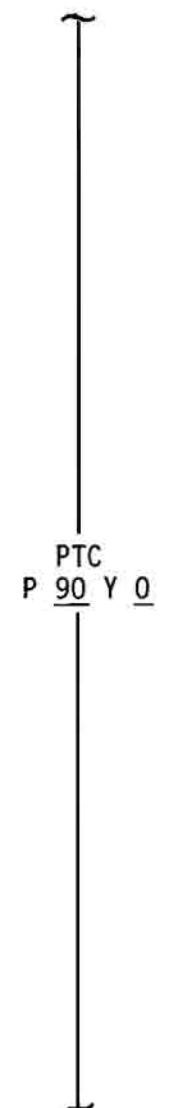
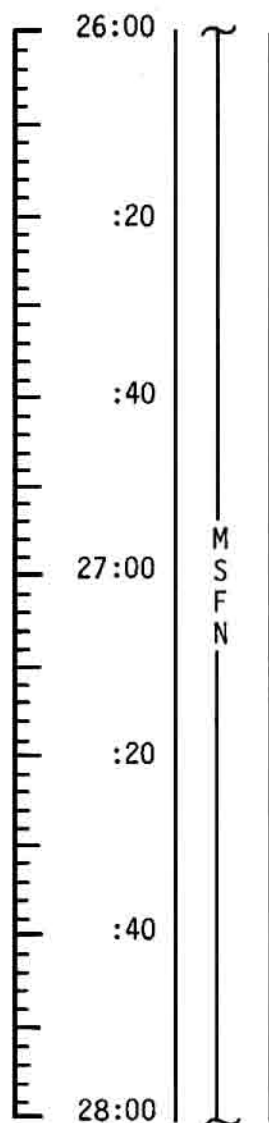
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	25:00 - 26:00	2/TLC	3-21

MCC-H

1513 CST

FLIGHT PLAN

NOTES



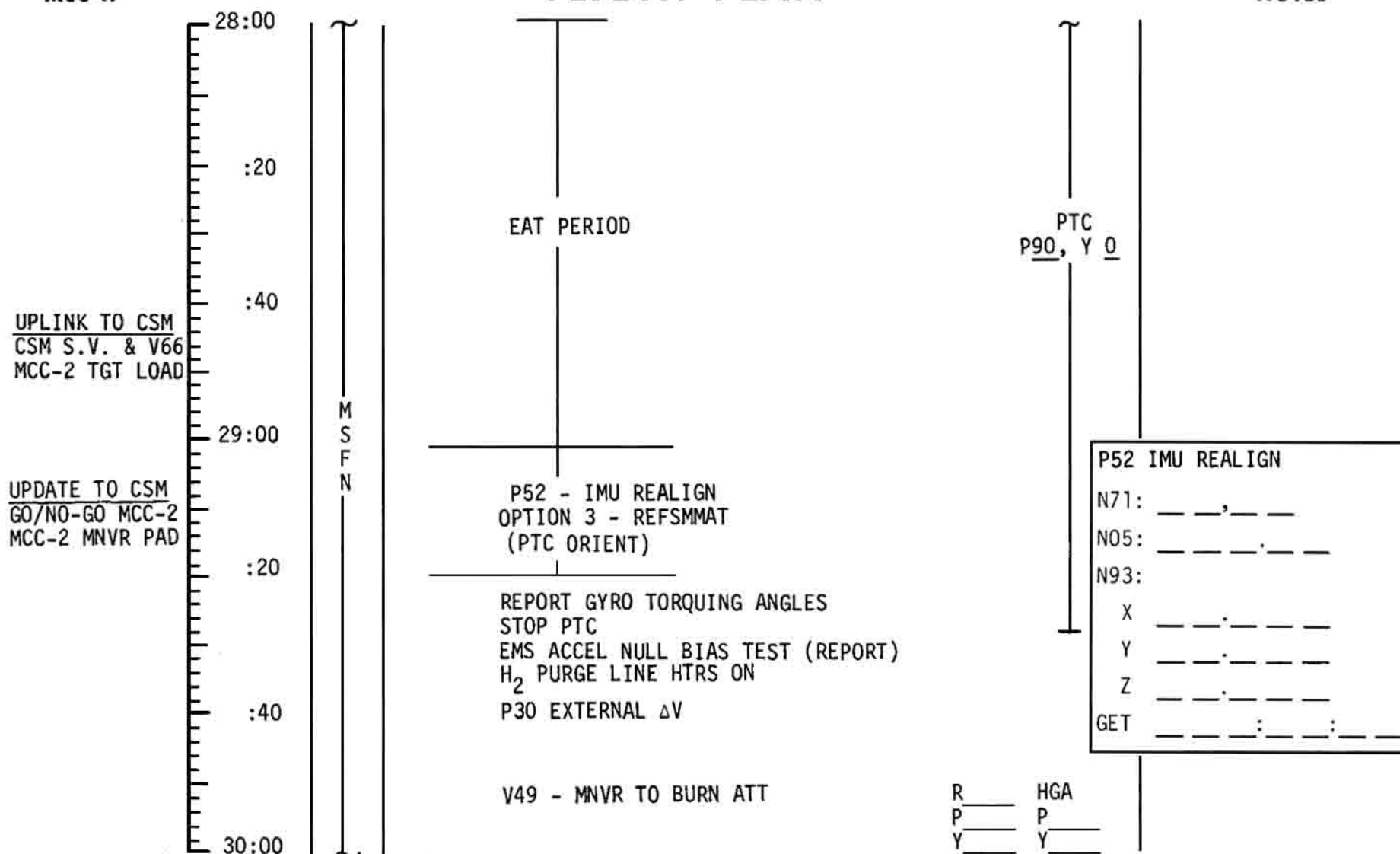
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	26:00 - 28:00	2/TLC	3-22

MCC-H

1713 CST

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	28:00 - 30:00	2/TLC	3-23

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FLIGHT PLAN

MCC-2
BURN TABLE

P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	IF<2FPS, TRIM X AXIS TO 0.2FPS IF>2FPS, NO TRIM

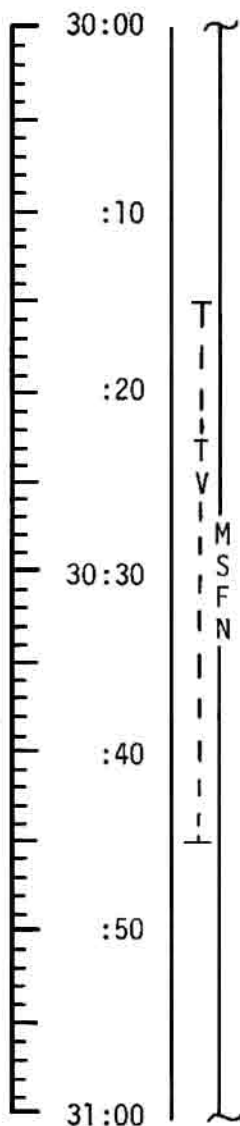
TABLE 3-3
3-23 A

1913 CST

FLIGHT PLAN

NOTES

TLI CUTOFF
+ 28 HRS



P40 - SPS THRUST
SXT STAR CHECK

TV (GDS) 30:15 TO 30:45
CM4/TV-AVG (f5.6)

H₂ & O₂ FUEL CELL PURGE
WASTE WATER DUMP
H₂ PURGE LINE HEATERS - OFF

GDC ALIGN

NOTE: MCC-2 WILL
BE ACCOMPLISHED
ON BANK A ONLY

MCC-2

V66 - TRANSFER CSM SV TO LM SLOT
MCC-2 BURN STATUS REPORT

TIG: 30:40:49
BT: 2.2 SEC
ΔVR: 15.1 FPS
ULLAGE: - NONE
ORBIT: N/A

BURN STATUS REPORT

X	X					ΔTIG
X	X					BT
						V _{gx}
X	X	X				R
X	X	X				P
X	X	X				Y
						V _{gx}
						V _{gy}
						V _{gz}
						ΔV _c *
X	X	X				FUEL *
X	X	X				OX *
X	X	X				UNBAL

*ITEMS TO BE REPORTED TO MSFN	
----------------------------------	--

ATTITUDE FOR MCC-2
BURN IS CONSTRAINED
IN ROLL FOR HGA
ACQUISITION FOR TV
AND BY SXT STAR CHECK

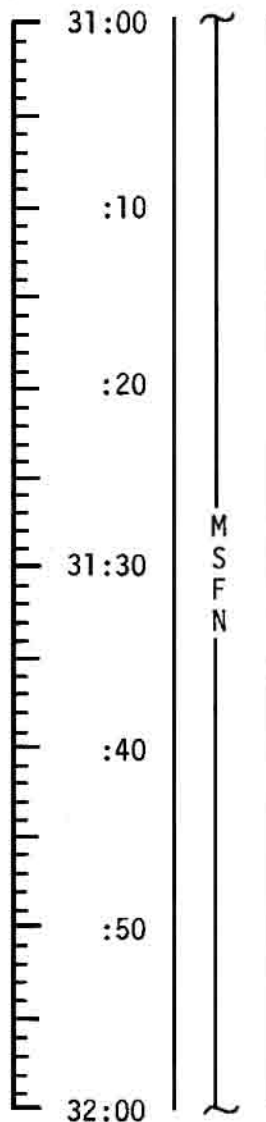
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	30:00 - 31:00	2/TLC	3-24

MCC-H

2013 CST

FLIGHT PLAN

NOTES



MNVR TO OPTICS CALIBRATION ATT R 147
 P23-CISLUNAR NAVIGATION P 340
 OPTICS CALIBRATION Y 0
 STAR 4 0

P00

V49 - MNVR TO SIGHTING ATT R 136
 STAR/EARTH HORIZON P 307
 P23 - CISLUNAR NAVIGATION Y 0
 LOAD W MATRIX (R1 + 4 5 0 0 0) (R2 + 0 0 0 0 6)

1. STAR 37 EFH (R3 00120) NUNKI

2. STAR 221 ENH (R3 00110) DELTA CAPRICORNI
 N88: (R1 + ~~03986~~) (R2 ~~48220~~) (R3 ~~-28249~~)

3. STAR ~~221~~ EFH (R3 00120) ~~DELTA CAPRICORNI~~ DELTA SAGITTAR II

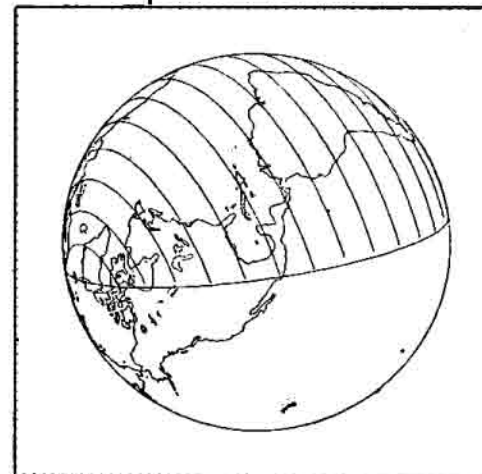
4. STAR ~~40~~ ENH (R3 00110) ~~DELTA CAPRICORNI~~ ENIF

5. STAR 77 EFH (R3 00120) KAUS AUST
 N88: (R1 +03986)(R2 -41062)(R3 -28249)

3 MARKS EACH STAR

INCORPORATE P23
 MARK DATA AND
 UPDATE ONBOARD
 STATE VECTOR

GET 30 HRS F.O.V. 4°



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	31:00 - 32:00	2/TLC	3-25

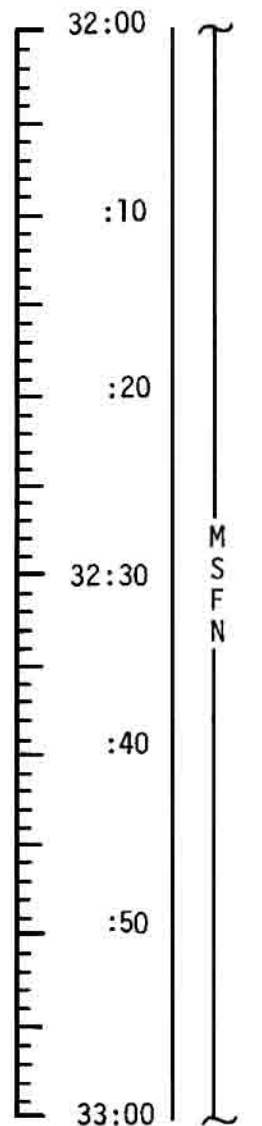
MCC-H

2113 CST

FLIGHT PLAN

NOTES

UPDATE TO CSM
QUADS TO DISABLE
FOR PTC (LOWEST
QUANTITY PRPLNT)



MANEUVER TO PTC ATTITUDE P 90
START PTC Y 0
S-BAND ANT - OMNI B ON MCC CUE
SECURE HGA
HGA TRACK - MAN
HGA PITCH -52°
HGA YAW 270°
CHECK BAT VENT (TEST METER 4A)

PTC
P 90 Y 0

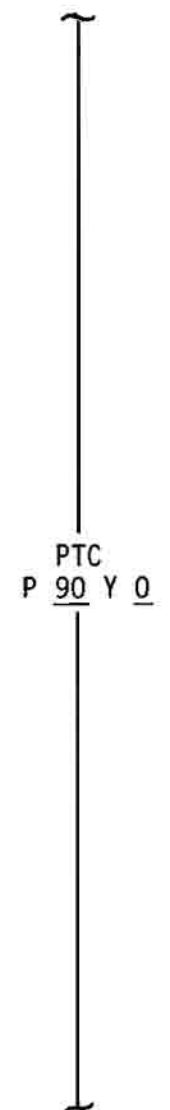
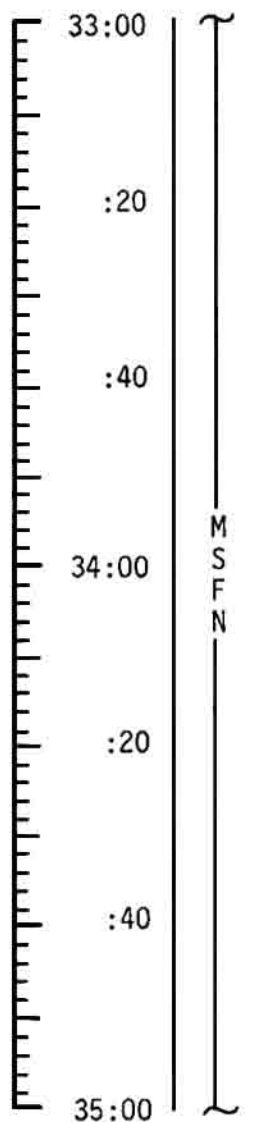
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	32:00 - 33:00	2/TLC	3-26

MCC-H

2213 CST

FLIGHT PLAN

NOTES



LOI MINUS 5 HR
FLYBY IS A
CIRCUMLUNAR
TRAJECTORY TO THE
PRI MPL AND
WITH A PERILUNE
BETWEEN 60 AND
1500 NM.

UPDATE TO CSM
LOI MINUS 5 HR
FLYBY

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	33:00 - 35:00	2/TLC	3-27

MCC-H

0013 CST

FLIGHT PLAN

NOTES

L10H CANISTER CHANGE
(5 INTO A, STOW 3 IN B5)

ONBOARD READOUT

BAT C _____

PYRO BAT A _____

PYRO BAT B _____

RCS A _____

B _____

C _____

D _____

DC IND SEL - MNA OR B

REPORT LM/CM ΔP

PTC
P 90 Y 0

EAT PERIOD

PRESLEEP CHECKLIST

UPLINK TO CSM
CSM S.V. & V66

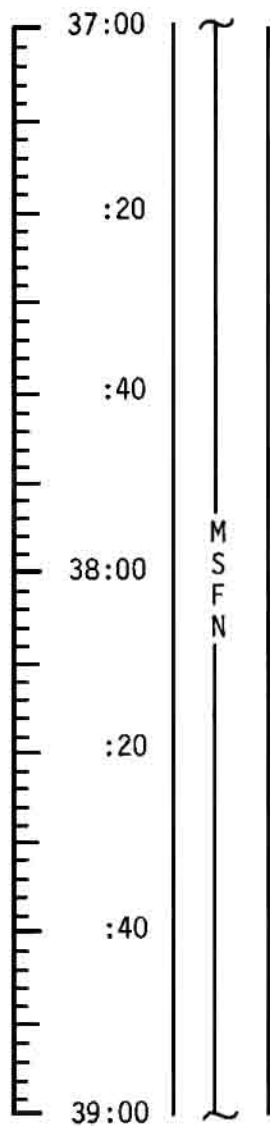
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	35:00 - 37:00	2/TLC	3-28

MCC-H

0213 CST

FLIGHT PLAN

NOTES



REST PERIOD
(10 HOURS)

PTC
P 90 Y 0

DURING REST PERIOD
TWO CREWMEN IN
REST STATIONS AND
ONE IN COUCH

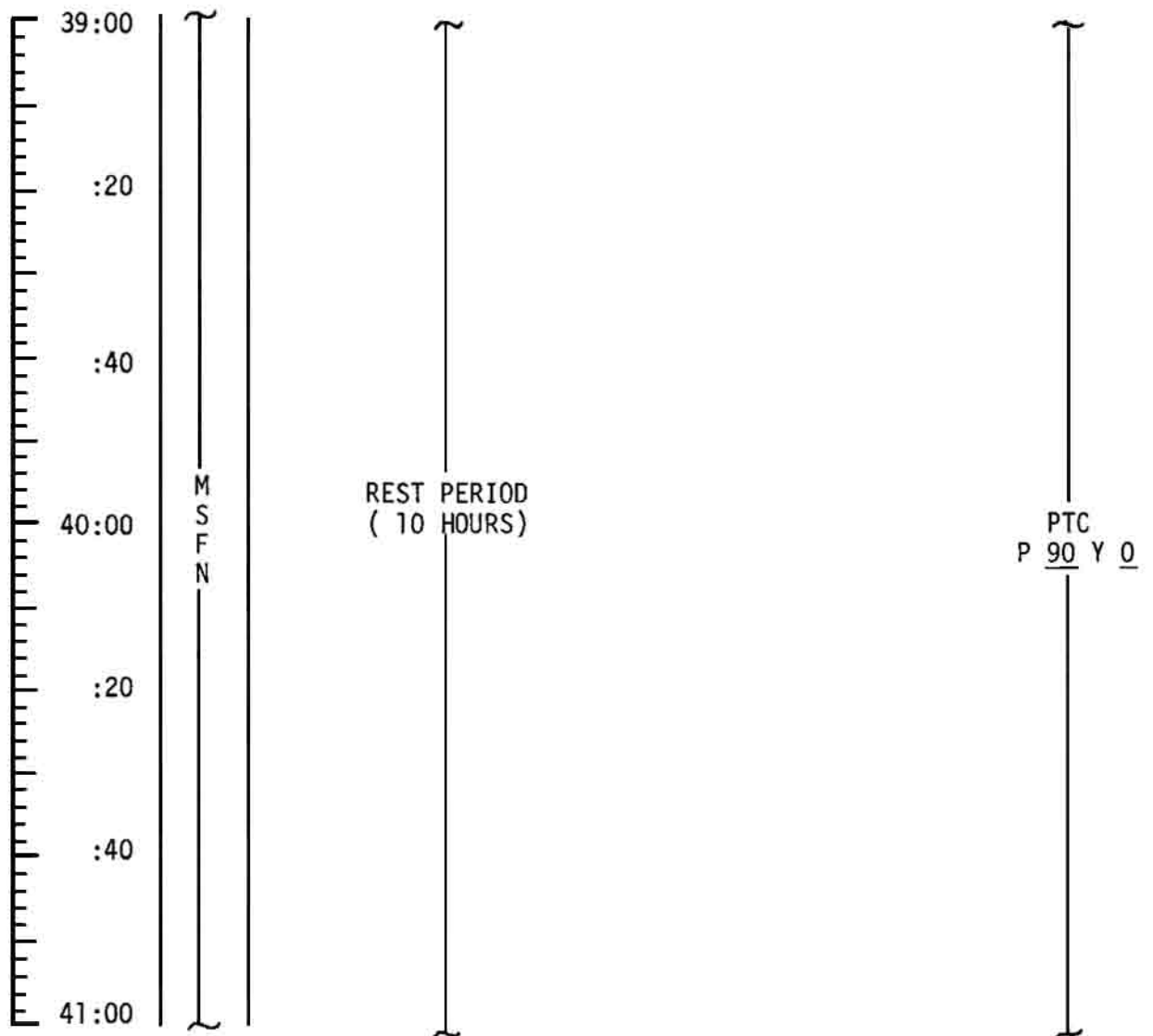
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	37:00 - 39:00	2/TLC	3-29

MCC-H

0413 CST

FLIGHT PLAN

NOTES



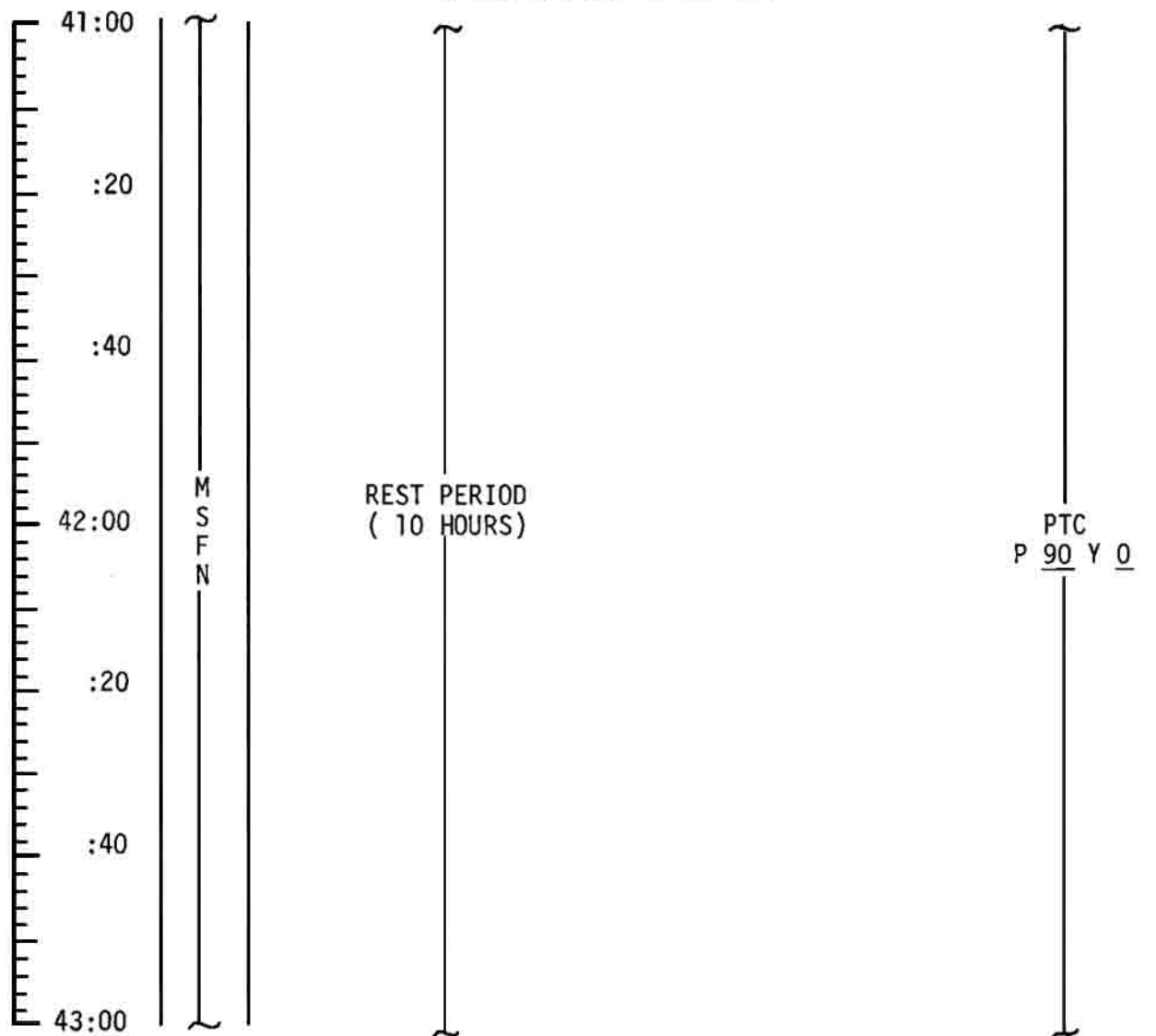
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	39:00 - 41:00	2/TLC	3-30

MCC-H

0613 CST

FLIGHT PLAN

NOTES



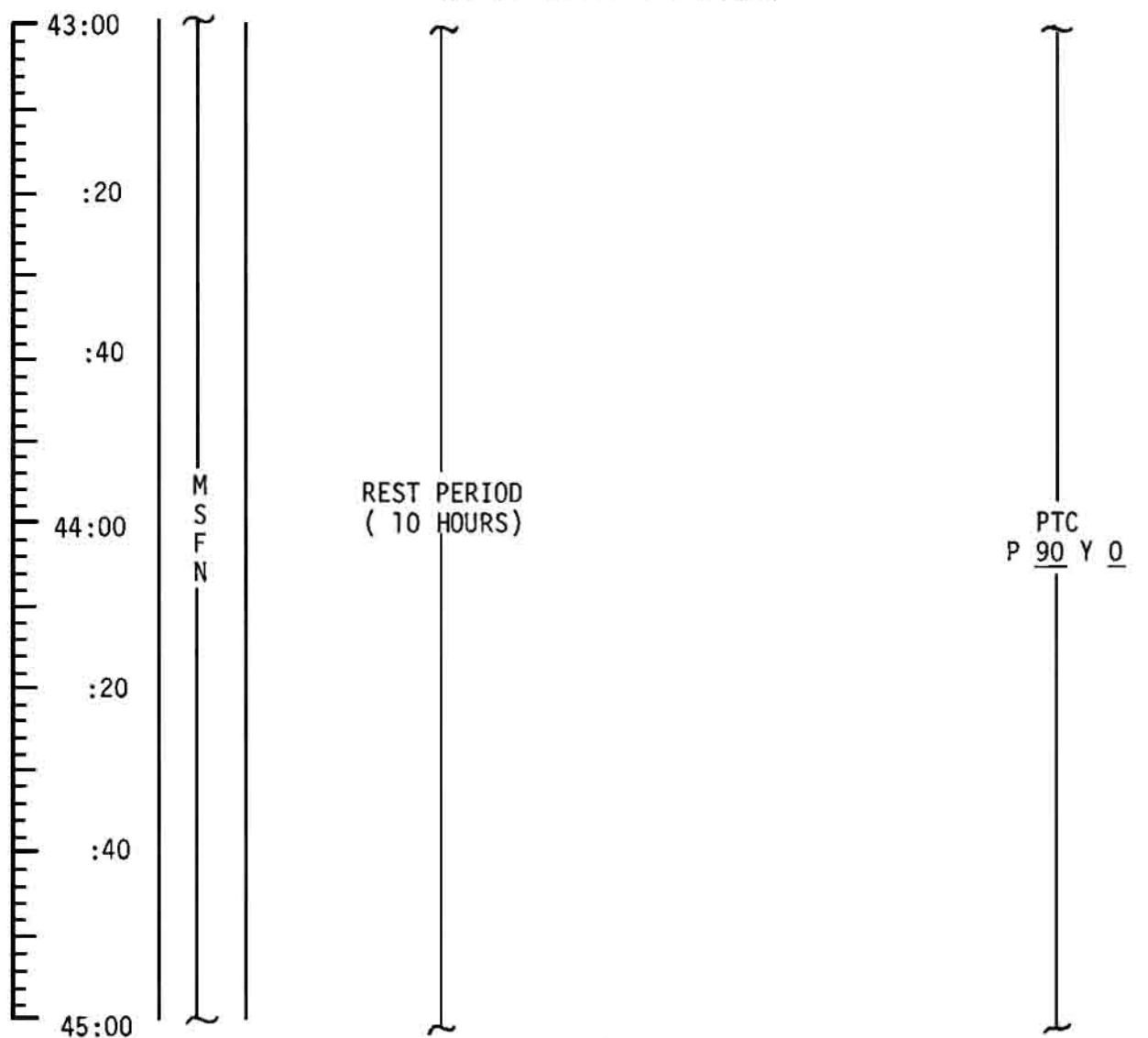
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	41:00 - 43:00	2/TLC	3-31

MCC-H

0813 CST

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	43:00 - 45:00	2/TLC	3-32

MCC-H

1013 CST

FLIGHT PLAN

NOTES

45:00
:20
:40
46:00
:20
:40
47:00

M
S
F
N

REST PERIOD
(10 HOURS)

PTC
P 90 Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16. 1970	45:00 - 47:00	2/TLC	3-33

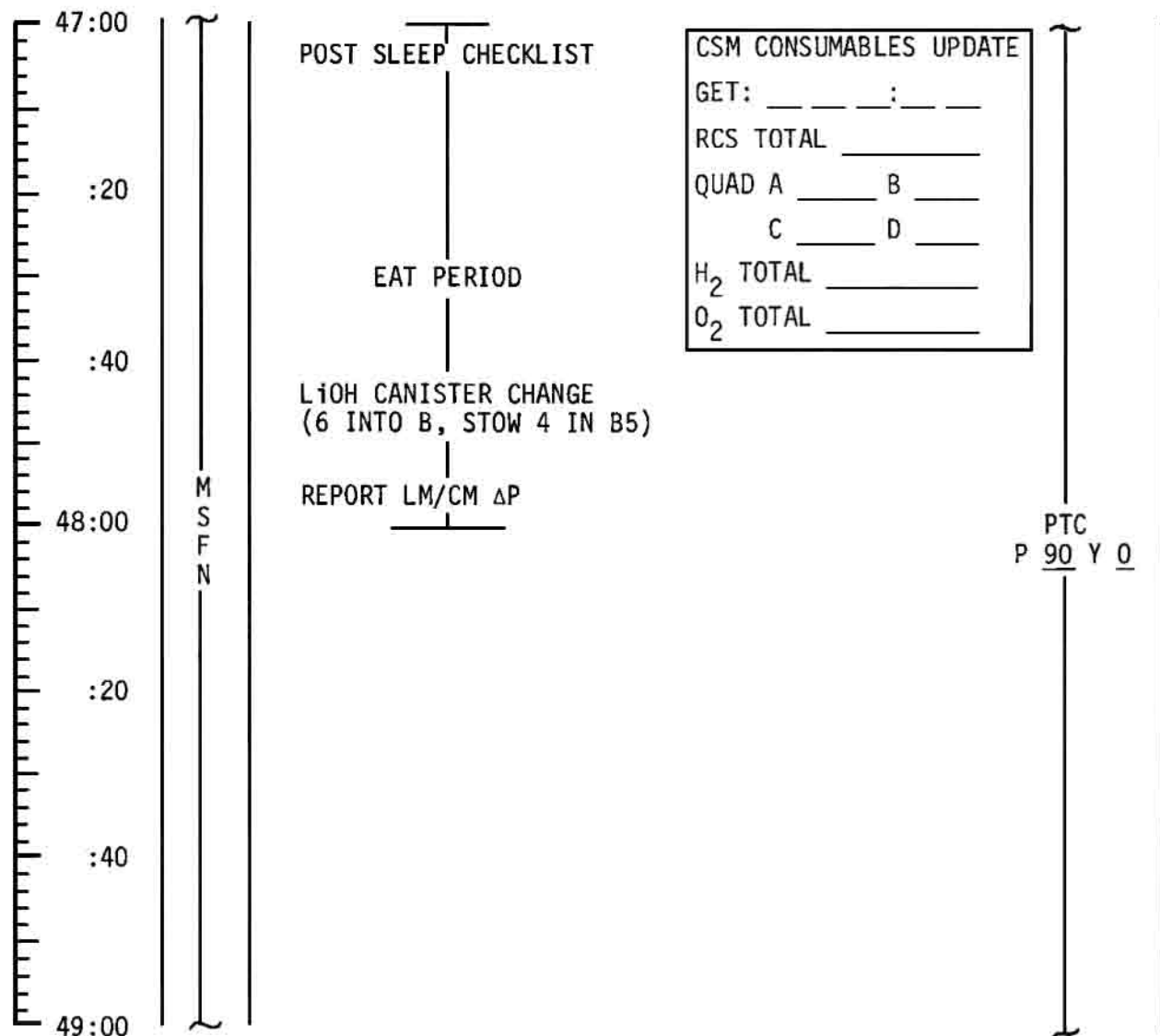
MCC-H

1213 CST

FLIGHT PLAN

NOTES

UPDATE TO CSM
CONSUMABLES
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	47:00 - 49:00	3/TLC	3-34

MCC-H

1413 CST

FLIGHT PLAN

NOTES

49:00

:20

:40

50:00

:20

:40

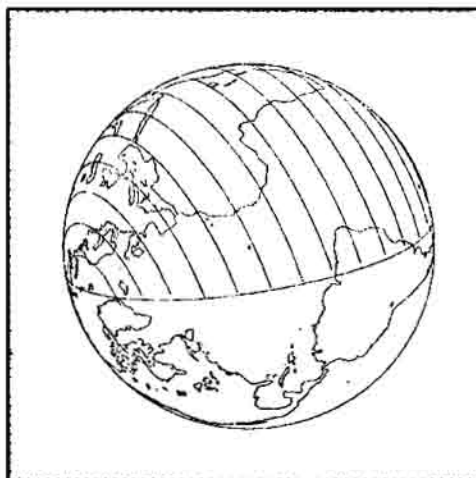
51:00

M
S
F
N

P52 - IMU REALIGN
OPTION 3 - REFSMMAT
(PTC ORIENT)

REPORT GYRO TORQUING ANGLES

GET 50 HRS F.O.V. 3°



P52 IMU REALIGN

N71: _ _ _ , _ _ _

N05: _ _ _ . _ _ _

N93:

X _ _ _ . _ _ _

Y _ _ _ . _ _ _

Z _ _ _ . _ _ _

GET _ _ _ : _ _ _ : _ _ _

PTC
P 90 Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	49:00 - 51:00	3/TLC	3-35

MCC-H

1613 CST

FLIGHT PLAN

NOTES

51:00
:20
:40
52:00
:20
:40
53:00

M
S
F
N

EAT PERIOD

PTC
P 90 Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	51:00 - 53:00	3/TLC	3-36

MCC-H

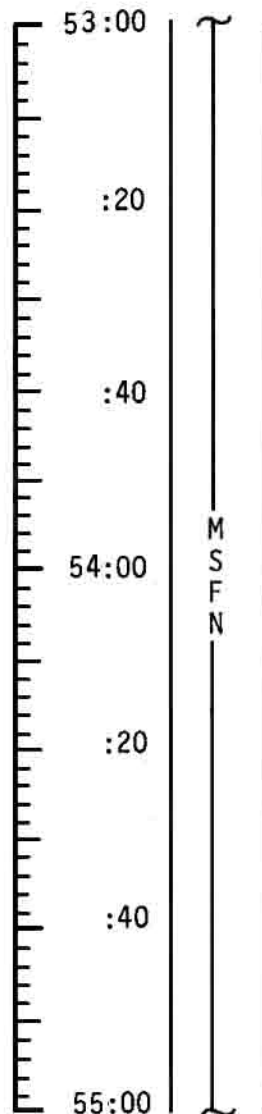
1813 CST

FLIGHT PLAN

NOTES

UPLINK TO CSM
CSM S.V. & V66
MCC-3 TGT LOAD

UPDATE TO CSM
GO/NO-GO MCC-3
MCC-3 MNVR PAD



P52 IMU REALIGN
OPTION 3 REFSMMAT
(PTC ORIENT)

CONTINUE PTC IF MCC-3 IS NOT PERFORMED
REPORT GYRO TORQUING ANGLES

P30 - EXTERNAL ΔV

PTC
P 90 Y 0

FUEL CELL PURGE AND
WASTE WATER DUMP
SCHEDULED AT 55:08
WILL BE DELAYED
TO 57:50 IF MCC-3
IS NOT PERFORMED

P52 IMU REALIGN

N71: _ _ , _ _

N05: _ _ . _ _

N93:

X _ _ . _ _

Y _ _ . _ _

Z _ _ . _ _

GET _ _ : _ _ : _ _

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	53:00 - 55:00	3/TLC	3-37

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FLIGHT PLAN

MCC-3
BURN TABLE

P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	IF<2FPS, TRIM X AXIS TO 0.2FPS IF>2FPS, NO TRIM

TABLE 3-4
3-37A

2013 CST

FLIGHT PLAN

NOTES

MCC-3 WILL BE
DELAYED TO MCC-4
IF PROPELLANT
COST IS NOT
PROHIBITIVE

BURN STATUS REPORT

X	X			•			ΔTIG
X	X			•			BT
							V _{gx}
TRIM					•		
X	X	X					R
X	X	X					P
X	X	X					Y
						•	V _{gx}
						•	V _{gy}
						•	V _{gz}
						•	ΔV _c
X	X	X					FUEL
X	X	X					OX *
X	X	X					UNBAL

*ITEMS TO BE
REPORTED TO MSFN

(LOI - 22 HRS)

UPDATE TO CSM
QUADS TO DISABLE
FOR PTC (LOWEST)
QUANTITY PRPLNT)

V49 - MNVR TO BURN ATT
P40/41 - SPS/RCS THRUST
SXT STAR CHECK
O₂ FUEL CELL PURGE
WASTE WATER DUMP

GDC ALIGN

TIG: 55:24:53
BT: NOM. ZERO
ΔVR: NOM ZERO
ULLAGE: NONE
ORBIT: N/A

V66 - TRANSFER CSM SV TO LM SLOT
MCC-3 BURN STATUS REPORT

BATTERY CHARGE, BATTERY B

START PTC

LM TUNL VENT VALVE - LM/CM ΔP
IF LM/CM $\Delta P < 1.7$ psid - VENT
UNTIL $\Delta P > 1.7$

PTC
P 90, Y 0

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	55:00 - 56:00	3/TLC	3-38

MCC-H

2113 CST

FLIGHT PLAN

NOTES

56:00
:10
:20
56:30
:40
:50
57:00

M
S
F
N

EAT PERIOD

PTC
P 90 Y 0

LUNAR PHOTOGRAPHY
AT CREW OPTION
CM/EL/80 or 250/BW
RING(f5.6.250.-)10
MAG P
CM/EL/80 or 250/CEX-
RING (f5.6.250.-)10
MAG L

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	56:00 - 57:00	3/TLC	3-39

MCC-H

2213 CST

FLIGHT PLAN

NOTES

UPLINK TO CSM
 ΔH (IF REQUIRED)

57:00

:10

:20

57:30

:40

:50

58:00

M
S
F
N

PRESSURIZE CSM TO 5.7 PSIA
PRESSURIZE LM

T EPHEM: V05N01 1706E
COPY IN LM ACTIVATION CHECKLIST (ACT 33)

CLEAR TUNNEL OF
CM HATCH
INSPECT TUNNEL &
DOCKING LATCHES
REMOVE PROBE & DROGUE

TV (GDS) 58:00 TO 58:30
CM__/TV-AVG (f5.6)

LOAD DAP. N46 (21111, 11111)

STOP PTC ROLL AT ____.

HGA P__ Y__

PTC
P 90 Y 0

ΔH DETERMINED
FROM STAR/EARTH
HORIZON SIGHTINGS
WILL BE UPLINKED
IF IT DIFFERS FROM
 ΔH IN E-MEMORY
BY MORE THAN 5.0 KM

O2 FUEL CELL PURGE
& WASTE WATER DUMP
@ 57:50 IF NOT
PERFORMED AT 55:08

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	57:00 - 58:00	3/TLC	3-40

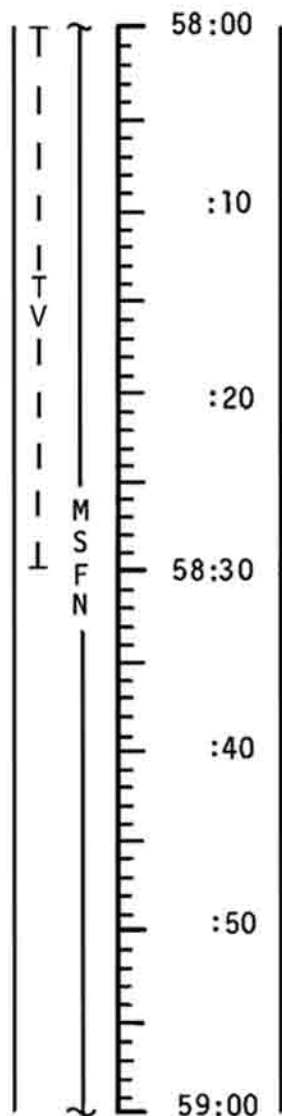
FLIGHT PLAN

CSM

CMP

TEMPORARILY STOW
PROBE & DROGUE

2313 CST



LM

CDR

ACTIVATION CHECKLIST

LMP

OPEN LM HATCH
RECORD AND REPORT ROLL
CALL ANGLE
IVT TO LM

IVT TO LM

ASSIST CDR

LM
FAMILIARIZATION

LM
FAMILIARIZATION

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	58:00 - 59:00	3/TLC	3-41

FLIGHT PLAN

CSM
CMP

MNVR TO PTC ATTITUDE
START PTC
S-BAND ANT - OMNI B
(ON MCC CUE)
SECURE HGA

P 90
Y 0

PTC
P 90 Y 0

M
S
F
N



LM

CDR

LM
FAMILIARIZATION

LMP

LM
FAMILIARIZATION

IVT TO CSM

IVT TO CSM

CLOSE LM HATCH

FLIGHT PLAN

CMP: INSTALL PROBE AND DROGUE
INSTALL CM HATCH
LM TUNNEL VENT VALVE - LM/CM ΔP

MCC-H

UPDATE TO CSM
QUADS TO DISABLE
FOR PTC (LOWEST
QUANTITY PRPLNT)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	59:00 - 60:00	3/TLC	3-42